Erik Löffelholz

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PROFILE

Software developer and computational scientist with a Master's degree in Mathematical Physics. Strong background in machine learning (PyTorch), numerical simulation, and full-stack web development. Experienced in building differentiable simulation systems, GPU-accelerated compute pipelines, and interactive 3D visualization tools. Combines deep mathematical foundations with hands-on engineering skills in Python, TypeScript, and modern web frameworks, and increasingly works with agentic coding models and LLM-assisted development workflows.

EDUCATION

Oct 2021 – Sep 2025

M.Sc. Mathematical Physics
Universität Leipzig
Focus: Differential Geometry · PDE Theory · Quantum Field Theory · Numerical Methods

Oct 2018 – Mar 2022

B.Sc. Physics
Universität Leipzig

PROFESSIONAL EXPERIENCE

Apr 2025 – May 2026

Scientific Software Developer

Max Planck Institute for Mathematics in the Sciences, Leipzig

Promoted from student developer to Scientist for the final period (Apr–May 2026).

- Built GPU-accelerated differentiable simulation and 3D visualization software (PyTorch, Three.js, WebGL) for mathematical research
- Implemented a discrete differential geometry operator library and pipelines for embedding meshes into curved (hyperbolic/spherical) spaces via energy minimization
- Designed a binary WebSocket streaming protocol (10–20× faster than JSON) connecting Python/PyTorch compute to browser frontends, with CUDA/MPS/CPU backends
- Co-authored a research paper on hyperbolic surface mesh embeddings (Bridges Conference 2026, under review)

2025 – Present

Independent Developer & Researcher

Graph ML, Differentiable Physics & Scientific Computing

- Built full-stack ML and simulation applications (PyTorch + FastAPI + React/Three.js): a from-scratch graph ML framework, a PINN solver for ten PDEs, and browser-based differentiable-physics demos
- Developed hands-on expertise with agentic coding models and LLM-assisted development workflows

2025 – Present

Subject-Matter Expert — STEM & Coding

Outlier.ai — AI Training & Model Evaluation

- Authored and reviewed STEM, mathematics and physics tasks for large-language-model training
- Evaluated and ranked model outputs (RLHF) and completed coding tasks in Python, C++ and JavaScript
- Contributed German-language (de-DE) prompt and audio-prompt tasks

Mar 2024 – Feb 2025

Working Student — Mathematics Editorial

Ernst Klett Verlag GmbH, Leipzig

- Reviewed and quality-assured mathematical content for educational textbooks
- Ensured technical correctness and pedagogical clarity across multiple publications

TECHNICAL SKILLS

Languages

Python · TypeScript · JavaScript · C++ · HTML/CSS

Machine Learning

PyTorch · Graph Neural Networks · Diffusion Models · VAEs · Neural ODEs · PINNs · LLM Evaluation (RLHF)

Web & Frontend

React · Three.js · WebGL/WebGPU · Vite · Tailwind CSS · WebSocket

Backend & Infrastructure

FastAPI · REST APIs · Docker · Git · CI/CD · Agentic Coding Tools

Scientific Computing

NumPy · SciPy · SymPy · Numerical PDE solvers · Monte Carlo methods

Spoken Languages

German (native) · English (C1 — full professional proficiency)

SELECTED PROJECTS

Mesh Embeddings & DDG — GPU mesh embedding into hyperbolic and spherical spaces via spring–mass energy minimization, with a discrete differential geometry operator library. Computational basis for the Bridges 2026 paper.

Graph ML Lab — From-scratch PyTorch framework for spatial graph generation: GCN, GAT, graph VAE, and joint discrete-continuous diffusion models over 3D graph structures. No external GNN libraries. github.com/erik2810/ml-projects

Mesh-Based Physics Simulator — Fully differentiable physics engine in PyTorch. Maps mesh topology to particle-spring systems with energy-based force computation and end-to-end backpropagation through dynamics.

PINN Solver — Physics-Informed Neural Network solver for 10 PDEs (Burgers, Heat, Wave, KdV, etc.) with PyTorch training backend and client-side JavaScript inference. github.com/erik2810/pde-solver

DiffQFT — Differentiable quantum field theory computations in AdS_2. Neural surrogates replacing Monte Carlo integration, PINN solvers, FastAPI backend. github.com/erik2810/DiffQFT

Knitted Models — Computational geometry engine for generating woven strand patterns on quad meshes using half-edge data structures and spline interpolation. Three.js rendering with GPU path tracing.

Neural ODE Engine — Neural network learning chaotic Lorenz dynamics in real time. From-scratch backpropagation, RK4 integrator, and Adam optimizer in pure JavaScript. github.com/erik2810/differentiable-physics-engine

Full portfolio with live demos at erik2810.github.io/projects

REFERENCES

Available upon request.